



Ultrasound measurements reveal predictable relationship between spawning time and ovarian growth in *Octopus vulgaris*

Lorenzo Márquez^{a,1}, Pauline Weterings^{b,1}, Samar Saber^c, Beatriz C. Felipe^d, Sofia Maccuro^b, Marta Serrano^d, Eve Seuntjens^b, Francisco Acosta-Cifuentes^c, María González^c, Eduardo Almansa^{d,*}

^a Centro de Investigación, Innovación y Creación (CIIC-UCT), NIPA, Facultad de Recursos Naturales, Universidad Católica de Temuco, Temuco, Chile

^b Laboratory of Developmental Neurobiology, Department of Biology, KU Leuven, Leuven, Belgium

^c Centro Oceanográfico de Málaga (COMA-IEO), CSIC, Málaga, Spain

^d Centro Oceanográfico de Canarias (COC-IEO), CSIC, Santa Cruz de Tenerife, Spain

ARTICLE INFO

Keywords:

Ultrasonography
Ovarian volume
Gonadosomatic index
Time to spawning
Specific growth rate
Octopus vulgaris females

ABSTRACT

There has been a great interest in common octopus *Octopus vulgaris* aquaculture in recent years and commercial production is within reach. *O. vulgaris* is a short-lived semelparous species with a long spawning period. Therefore, monitoring ovary maturation in cultured octopuses can be useful in the scheduling of hatchery operations and management of culture resources. The present work assessed ovary volume as a proxy for maturity status in cultured and wild common octopus by means of a noninvasive technique, ultrasonography, to anticipate the time of spawning in aquaculture. A total of 27 octopuses (15 cultured females with initial wet weights from 600 to 4600 g, and 12 wild females weighing 990–4800 g) in different ovarian maturation stages were used in ultrasonographic procedures. Ultrasound-based ovary volumes were compared with real ovarian volumes and then the ultrasonography-based gonadosomatic index (*GSIU*) was estimated. Data from cultured females indicate that *GSIU* was initially below 0.3 cm³ per 100 g of body weight in immature animals, and attained a fast increase until reaching about 4.0 cm³ per 100 g of body weight just before spawning. The steep increase in relative volume occurred similarly in females of different body weights. Furthermore, the specific growth rate of cultured animals decreases with time, and with *GSIU*. Lastly, a linear mixed model indicated a significantly negative and logarithmic relationship between the time to spawning and *GSIU* in captive females. This relationship can be used to predict spawning time in immature and maturing females, helping in the management of octopus broodstock in research and production facilities.

1. Introduction

The common octopus *Octopus vulgaris* Cuvier, 1797, is an important commercial cephalopod, and it is presently under intensive research to develop its industrial production. The lifespan of this semelparous species is estimated to be between twelve and fifteen months (Katsanevakis and Verriopoulos, 2006). Juvenile specimens of the common octopus easily acclimatise to farming conditions and can attain fast growth (Sánchez et al., 2014), and the survival and growth of the paralarval stage have recently been improved to values suitable for commercial production (Almansa et al., in press). In addition, the manipulation of the duration of embryonic development, usually taking several weeks,

helps researchers and potential farmers to manage resources for larval culture (Márquez et al., 2023). However, no similar knowledge is available to predict the spawning time of octopus females. The prediction of embryonic development duration, and the estimation of female spawning time would be particularly useful to schedule the use of tanks and other resources for the culture of octopus early stages. One potential way to produce suitable estimations of the spawning time is by monitoring the ovarian maturation. The gonad maturation in the genus *Octopus* is influenced by environmental factors such as temperature, photoperiod and food availability (Forsythe and Hanlon, 1988; Mangold, 1983). At the same time, maturation is regulated by neuro-endocrine secretions that integrate the effects of external factors

* Corresponding author.

E-mail address: eduardo.almansa@ieo.csic.es (E. Almansa).

¹ Both authors share first authorship

(Polese et al., 2015; Di Cristo, 2013; Wells and Wells, 1959). Regardless of the factors shaping the process, the stage of maturation in wild females can be assessed by calculating the gonadosomatic index (GSI), which is a measure of the relative volume of the gonad (Sieiro et al., 2020, 2014). As such, the GSI can also be an indicator of the reproductive status in cultured specimens of *O. vulgaris*, particularly if the procedure does not require animal slaughtering, as with biopsy or ultrasonographic techniques.

Considering the research on fish reproduction as a reference, ovarian biopsy (Du et al., 2017; Coward and Bromage, 2001), blood sex hormones (Du et al., 2017; Mlingi et al., 2023; Evans et al., 2004), and ultrasonographic imaging (Novelo and Tiersch, 2012) or, more recently, machine-learning-guided interpretation of ultrasonographic imagery (Yari et al., 2024), are all non-destructive methods chosen for assessing the maturation status. Among these non-destructive procedures, ultrasonography has the advantage of being a non-invasive technique. Ultrasonod images of gonads of fish and gastropod females are suitable for measuring primary biometric variables such as ovary cross-section area, circumference and diameter (McGarvey et al., 2021; Naeve et al., 2018; Jennings et al., 2005; Mattson, 1991), and ovary antero-posterior length (Naeve et al., 2018; Whiteman et al., 2005). Some authors provide derived biometric variables such as ovary volume, which requires a model for ovary shape (Bryant et al., 2007; Bureau du Colombier et al., 2015; Dasgupta et al., 2024) or machine-learning techniques (Yari et al., 2024) and, subsequently, volumetric estimations of the GSI (Dasgupta et al., 2024). In general, ultrasonographic variables of fish ovaries show a good correlation with the same variables measured in the dissected organ (Naeve et al., 2018; Mattson, 1991).

In relation to potentially useful non-destructive methods to study the maturation process in cephalopods, several facts should be taken into account: i) there is a lack of publications about safe procedures to implement needle biopsy (Ponte et al., 2017), ii) the chemical nature of sex steroids is still under active research (Wang et al., 2022; Minakata and Tsuitsui, 2016), and iii) ultrasound imagery with anaesthesia is classified as a mild research procedure (Cooke et al., 2019), and has been used to study digestive tract, brain and arm tissue architecture of cephalopod mollusks (Ponte et al., 2017; Berrueta et al., 2025; Margheri et al., 2011; Grimaldi et al., 2007), as well as reproductive condition in gastropods mollusks (Boles et al., 2023; Zou et al., 2024).

The present work aims to characterize gonadal maturation in *O. vulgaris* females to anticipate the time of spawning with application to aquaculture and, also, to manage the broodstock. Ultrasonography has been evaluated as a technique to monitor temporal changes in gonad size, and to estimate an ultrasonography-based volumetric gonadosomatic index (GSIU). These data allowed us to investigate the relationships between GSIU and the time to spawning (t_{spawn}), as well as the specific growth rate (SGR) in cultured female specimens.

2. Material and methods

2.1. Ethical approvals for cultured and wild animals

All animal experiments with cultured animals were performed in accordance with Spanish law 53/2013 in the framework of the European Union directive on animal welfare (Directive 2010/63/EU) for the protection of animals used for scientific purposes, following the Guidelines for the care and welfare of cephalopods in research proposed by Fiorito et al. (2015). The described study was approved by the Ethics Committee of the National Competent Authority (Canary government, project number CEIBA 1377–2023 and CEIBA 1610–2024). Ethical approval was not required for protocols with wild animals, as all wild animals in this study originated from commercial fisheries.

2.2. General experimental design

The general experimental design consists of the following two

phases:

- i) Obtaining a correlation between ovary volume determined by ultrasonography imaging and real ovary volume in wild females.
- ii) Monitoring the temporal change of the volumetric gonadosomatic index of captive females by ultrasonography (GSIU) until the spawning time. Subsequently, the relationship between the time to spawning and GSIU was explored, taking into account that the whole set of data must be grouped by female. The obtained regression line is proposed to be used to estimate the spawning times in *O. vulgaris* females similarly reared in captivity, provided that its GSIU is known.

2.3. Sample collection, sampling area and maintenance in captivity

Cultured animals. A total of 15 females were caught by artisanal fishery (trap gear) in Tenerife waters (Canary Islands) from September 2023 to September 2024. The initial weight of the captive octopuses ranged between 600 and 4600 g, with a median wet weight close to 1600 g. Culture time (from the first wet weight measurement until spawning) was between 54 and 163 days. Thirty-seven estimations of ovary volume by ultrasonography were obtained throughout a period of approximately 1.5 years. These individuals were maintained in 4000 L rectangular fiber-glass tanks at the Tenerife Culture Plant (IEO-CSIC) following the conditions described by Reis et al. (2021). The tanks were maintained under a natural photoperiod for the geographical location (between 10 L:14D to 14 L:10D). Temperature and salinity were $22.6 \pm 1.1^\circ\text{C}$ and 36.8 ± 0.1 PSU, respectively. Temperature was measured with a Tinytag Plus 2 thermometer (TGP-4020; Gemini Data Loggers Ltd., Chichester, West Sussex, United Kingdom) and salinity with a Refractometer S/Mill-E (ATAGO). About 50 % of each tank surface was covered with a shady green net. The water flow of the rearing tank was 5 L/min. A mixture of frozen food based on squid (*Loligo gahi*) and crab (*Callinectes sapidus*) were given *ad libitum* to the broodstock each day. PVC pipes and plastic boxes were placed inside the tanks to provide 2–3 dens for each specimen. The presence of eggs was verified on a weekly basis to avoid disturbing the breeders.

Wild animals. A total of 51 wild females were obtained from landings from artisanal (pots) and trawl fisheries in the Mediterranean Andalusian waters, between March and May 2024. Fresh octopuses were analyzed in a laboratory where the following measurements were recorded: total weight (TW) (to the nearest 0.1 g), dorsal mantle length (to the nearest mm), ovary weight (to the nearest 0.01 g), and ovary volume (to the nearest cm^3) measured by the water-displacement method according to Scherle (1970) and Saber et al. (2015). A total of 12 out of the 51 females (Supplementary Table S1) were subjected to ultrasonographic procedures to obtain ultrasound-based estimations of ovary volume (see below). Wet weights (WW) of wild octopuses included in the regression study were between 990 and 4800 g, with a median weight of approximately 1700 g. Ovary maturation stages ranged from stage 1 to stage 3 A according to the Medits scale (Medits Working Group, 2017).

2.4. Anaesthesia procedure for cultured common octopus

The common octopus was gently removed from its home tank using a net, subsequently transferred to a vessel with a secure lid, as they often try to escape, and submerged in aerated seawater containing 1.5 – 2 % absolute ethanol (EtOH) (PanReac, AppliChem) which is an effective anaesthetic for *O. vulgaris* according to Fiorito et al. (2015). During each sampling procedure, the anaesthetized individual was held and the ovary was scanned and the TW was recorded (to the nearest 0.1 g). The octopuses were then immediately returned to water, where their recovery was monitored.

2.5. Ultrasonographic procedures, ovary volume, gonadosomatic index, and specific growth rate in cultured female octopuses

In order to obtain a scan, the octopus was positioned on its ventral side and an 8 Mhz linear echo sounder (Esaote Pie Medical model, Falco 100) was used. Ultrasonography-based ovarian volume was estimated using the biometric measurements obtained from ultrasound images (anteroposterior, dorsoventral and left-right axes), and the *GSIU* was subsequently calculated considering the ovary as an ellipsoid ($Vol_U = (4/3) \pi r_1 r_2 r_3$, where r_1 , r_2 and r_3 are the three semi-axes). Female total body weight (*TW*) was also measured to the nearest 0.1 g. A volumetric *GSIU* was calculated for cultured females of common octopus as: $GSIU \text{ (cm}^3 \text{ per 100 g)} = 100 \times Vol_U \text{ (cm}^3 \text{)}/TW \text{ (g)}$. In addition, the *SGR* was calculated as the percentage of *TW* increase per day: $SGR = (\ln TW_2 - \ln TW_1)/\text{time}$, where TW_1 and TW_2 are the initial and final total weights for the period under consideration.

2.6. Ultrasonographic procedures, ovary volume and gonadosomatic index in wild female octopuses

Ultrasonographic procedures in wild specimens from the Mediterranean coast were performed with an ultrasound equipment (Mindray M9 VET) and a L12-4S linear array to scan ovaries in the same position as in the case of the cultured octopuses (Supplementary Figure S1). The frequency range was 6.6–12 Mhz.

2.7. Statistical analyses

The relationship between ultrasonographic ovarian volume and real ovarian volume in wild females was analysed by means of least squares linear regression. The point corresponding to the female with the largest volume was below the general regression line for the other 11 females. Since the ultrasonographic estimation of volumes of large organs can be limited by the size and type of the ultrasound transducer (McGarvey et al., 2021), this point was excluded from the regression.

The relationship between *GSIU* and time to spawning in captive females was analysed by means of a linear mixed model (LMM), with the individual as the cluster variable. Time to spawning was selected as a dependent variable and *GSIU* as covariate. A second LMM with time to spawning as dependent variable and $\ln(GSIU)$ as covariate was also run. For each LMM, intercept, slope, or intercept and slope were tested as random effects. The selected model was that one resulting in a non-singular solution and a lower Akaike Information Coefficient (AIC).

In order to study the relationship between the *SGR* and *GSIU*, *SGR* values were plotted against the mean *GSIU* obtained in the same period used to calculate the *SGR*. When a LMM was fitted, only singular solutions were obtained, thus, this type of analysis was ruled out. As an alternative, the ratio between the increment in *SGR* (between the last and the first measurement), and the increment in *GSIU* (also between the last and the first measurement) was calculated for each individual. This ratio was calculated in individuals with two or more *SGR* estimations, and was subjected to a one-sample *t*-test against a zero value. Statistical significance was set at 0.05 in all tests.

3. Results

3.1. Relationship between the ultrasonography-based ovarian volume and real ovarian volume in wild female common octopus

To benchmark ultrasonography as a valuable method to measure gonadal volume, ultrasonography measurements were first compared with actual volumetric measurements of dissected gonads obtained from *O. vulgaris* females from the wild. The following two estimations of ovary volume were obtained: one estimation based on ultrasound images, and the other one based on the water-displacement method of Scherle (1970). Both estimations of ovarian volume were compared by

regression techniques.

Fig. 1 shows the relationship between the ultrasonographic-based ovarian volume (Vol_U) (cm^3) and the Scherle-based ovarian volume ($Vol_{Scherle}$) (cm^3) in 12 wild females of *Octopus vulgaris*. When $Vol_{Scherle}$ was below 120 cm^3 , a linear correlation was obtained ($n = 11$, $R > 0.999$, $P < 0.001$), so that $Vol_U = 0.7408 Vol_{Scherle}$. As previously mentioned, the point with a $Vol_{Scherle}$ close to 180 cm^3 is clearly below the regression line, and was excluded (Fig. 1).

3.2. Relationship between ultrasonography-based gonadosomatic index (*GSIU*) and body wet weight (*WW*) in cultured females of *O. vulgaris*

Having established ultrasonography as a reliable method to measure gonadal size in a non-invasive manner, we next applied the method over a longer period of female maturation. The ultrasound images were used to monitor the temporal changes in ovary size of 15 cultured females. Ultrasonographic images of the ovaries were obtained every 15–30 days from capture to the time of spawning (period varying from a minimum of 26 days to a maximum of 101 days). This study combined animals of different sizes at the onset of measurements.

Ten out of 15 females underwent a fast increase in *GSIU*, starting when total weight (*TW*) was approximately between 2 and 3 kg. Three animals showed a similar rapid increase in *GSIU* when *TW* was larger than 4 kg. In addition, the rapid change in *GSIU* started at a weight below 2 kg in two females (Fig. 2). The starting value of *GSIU* was typically below $0.3 \text{ cm}^3 (100 \text{ g})^{-1}$. The last measured *GSIU* showed more variability, depending on the degree of gonad maturation, but the highest values were close to $4.0 \text{ cm}^3 (100 \text{ g})^{-1}$. These results show variability in the weight at which ovarian maturation begins, suggesting that maturation might be regulated independently from body weight.

3.3. Relationship between ultrasonography-based gonadosomatic index (*GSIU*) and time to spawning (t_{spawn}) in cultured females of *O. vulgaris*

As the end of maturation results in a spawning event, we hypothesized that *GSIU* measurements could provide a means to predict the spawning time. Plotting the *GSIU* values with time to spawning, it was observed that *GSIU* values increased from 0.1 to 0.2 cm^3 per 100 g *TW* for immature females, and to 4.0 cm^3 per 100 g *TW* in females on the verge of spawning. A negative correlation ($r = -0.85$, $P < 0.001$) was observed between *GSIU* and time to egg laying (Fig. 3). The whole set of points suggests a curvilinear and concave relationship.

The comparison of Akaike Information Coefficient (AIC) for linear mixed models with different random effects (slope, intercept, or slope plus intercept) indicated that a model with random intercept and fixed slope was the most suitable. The fixed effect (*GSIU*) was highly significant ($P < 0.001$), and it explained about 68 % of the variance of the dependent variable ($R^2 \text{ Marginal} = 0.684$), whereas the combination of fixed plus random effects explained more than 83 % of the variance (R^2

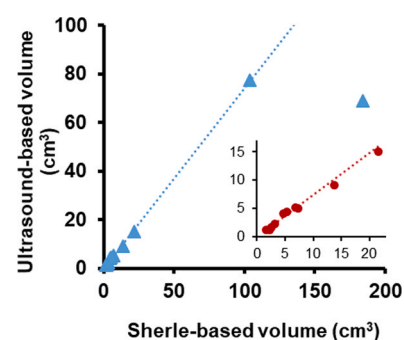


Fig. 1. Relationship between the ultrasound-based volume and the Scherle-based volume of the ovary in wild *Octopus vulgaris*. The inserted plot magnifies the regression line for Scherle-based volumes between 0 and 25 cm^3 .

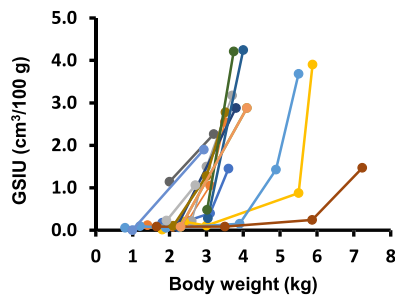


Fig. 2. Changes in ultrasonographic GSI with total weight in females of *Octopus vulgaris* reared in captivity. Different colours represent different individuals.

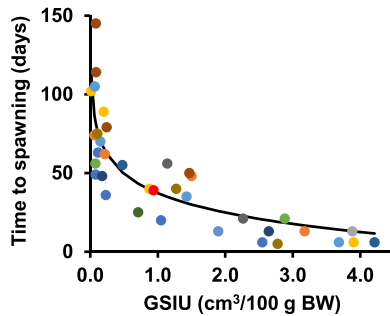


Fig. 3. Relationship between gonadosomatic index (GSI) and the number of days until the spawning event for each octopus. Different colours represent different individuals. The black line represents the regression line derived from a linear mixed model with a random intercept among individuals and fixed slope.

Conditional = 0.835). The mixed model resulted in the following equation to predict spawning time: $\text{Time to spawning (days)} = -17.8 \ln(\text{GSIU, cm}^3(100 \text{ g})^{-1}) + 37.2$. As shown in Fig. 3, the relationship was curvilinear and concave, with the slope being more negative for GSIU values between 0 and 1.0. However, it became nearly linear for GSIU above 1.0.

As shown in Fig. 3, the relationship was curvilinear and concave, with the slope being more negative for GSIU values between 0 and 1.0. However, it became nearly linear for GSIU above 1.0.

3.4. Relationship between specific growth rate and ultrasonographic index (GSIU) of cultured females of *Octopus vulgaris*

After obtaining the above data, we wanted to use this data to calculate to what extent gonadal maturation was correlated or not with body growth. The SGR between two consecutive ultrasonography sessions was calculated to monitor the specimen's growth rate up to the final session before spawning. The relationship between GSIU and SGR is shown in Fig. 4. It shows a decrease in SGR with ovary size. The linear mixed model equation was not fitted in this case due to the production of singular solutions. However, the ratio $\Delta \text{SGR} / \Delta \text{GSIU}$ was negative for all the individuals with more than one SGR estimation. The mean value of this ratio was -0.96 (Fig. 4b), and it was significantly different to zero in a t-Student test ($n = 8$, $t = -4.49$, $P = 0.003$). Thus, a negative relationship between SGR and GSIU was obtained, indicating that growth slowed down when animals approached spawning.

4. Discussion

The ultrasonography-based ovarian volume showed a good and positive correlation with the real ovarian volume of wild females. However, the regression line underestimated the real ovarian volume by approximately 25 %. Similarly, ultrasonography has also been reported

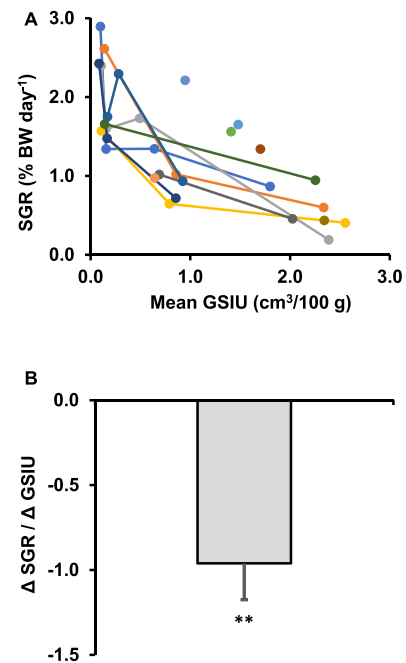


Fig. 4. Relationship between GSIU ($\text{cm}^3/100 \text{ g}$) and SGR (% BW/ day). A. Plot of SGR versus mean GSIU for each individual. B. Mean and standard error of the ratio $\Delta \text{SGR} / \Delta \text{GSIU}$. Different colours represent different individuals. Double asterisk indicates a significance level between 0.01 and 0.001.

to underestimate ovarian volume in fish while showing a good correlation between estimated and real volumes (Dasgupta et al., 2024; Bryant et al., 2007). Inaccuracies of the shape model selected for the calculation of ovarian volume can produce deviations from real values. Researchers studying fish gonads frequently assume fusiform-like models for the shape of fish ovaries, since they are typically much longer than wider (Bryant et al., 2007; Bureau du Colombier et al., 2015; Dasgupta et al., 2024; Macbeth et al., 2011). In the case of the common octopus, it is more reasonable to choose an ellipsoidal geometry as a model for ovary shape. However, when the ultrasound transducer is placed on the mantle, the ovary likely becomes flattened at the point of pressure, which implies that ovary shape would not exactly fit the ellipsoidal model during the ultrasonographic procedure. Ovary flattening due to transducer operation has already been reported in fish (McGarvey et al., 2021). In any event, it is clear that this effect does not necessarily affect the adequacy of the regression line, but only the value of the regression slope, thus it is not a serious problem since such a bias can be corrected.

Size estimation of large gonads by ultrasonography has been reported to result in underestimations depending on transducer type and size (McGarvey et al., 2021; Jennings et al., 2005). This effect can be a more relevant problem because the bias would only affect large gonads in relation to the transducer, potentially affecting the graphic pattern displayed by data sets. In any case, this bias is limited to the more upper range of gonad sizes. In the present work, one ovary with an estimated volume above 180 cm^3 stood out in the general regression line between ultrasonographic volume and real volume. More ultrasonographic measurements of octopus large ovaries (volume estimations clearly above 100 cm^3) will be useful to ascertain if this type of gonads is systematically below the regression line for smaller ovaries. Nevertheless, the data presented herein highlight the utility of ultrasonographic estimates for ovaries of at least up to 100 cm^3 with the linear transducer L12-4S (used in wild octopuses), and supports a simple equivalence between estimated and real ovarian volumes if an ellipsoidal shape model is assumed (Fig. 1): ultrasonographic volume = 0.74 (real volume). Taking into account this volume equivalence, together with the

mean ovary density found in wild females ($n = 47$, 1.108 ± 0.041), the weight-based *GSI* can be estimated as approximately 1.5 times ($1.108/0.74$) the ultrasonographic volume-based *GSI* (*GSIU*).

The most important finding of the present work is a predictable and quantifiable negative relationship between the time to spawning (the period between ultrasonography time and spawning time), and *GSIU*. The physiological basis for this relationship rests on the increase in the number of oocytes, and the increase in oocyte size during ovary maturation (Cuccu et al., 2013a). The linear mixed model presented herein indicates that time to spawning decreased from approximately 40 to 10 days when the *GSIU* progressed from 1.0 to 4.0 cm³ (100 g BW)⁻¹ or, after applying the 1.5 equivalence, when the real *GSI* ($GSI = 100 \times \text{ovary weight (g)}/\text{total weight (g)}$) increased from 1.5 to 6.0 g (100 g TW)⁻¹. The initial bent shape of the curve, and the large variation in time measurements when *GSIU* is small, probably stem from the grouping of pre-maturation females (that have small gonads for prolonged periods), and females that are entering the ovarian maturation. When gonads are small (pre-maturation), prediction becomes unreliable. Further research should be focused on the regression line for females with *GSIU* above 2.0 (real *GSI* approx. above 3.0), when females have embarked on the maturation process, since the relationship seems to be linear or quasi-linear above this value. Further refinement of the regression can make measurement even more reliable, and in theory should make prediction of spawning time feasible after a single measurement in this quasi-linear period.

A noteworthy finding was that maturation seemed to occur in a stereotyped manner, independent of the body weight of the female. Nearly all the females in the present work attained a fast increase in *GSI* at some point after being stocked in rearing tanks. This pattern with a fast change in *GSI* has been clearly obtained in wild females from the Tyrrhenian Sea during spring and summer, when the mean *WW* was above 1700 g, and after a peak of sex steroids in the ovary (Di Cosmo et al., 2001). Data on European populations of the common octopus support that the weight at which 50 % of females reach maturity (TW_{50}) ranges from 500 g to 1400 g in Mediterranean populations (Cuccu et al., 2013b; González et al., 2011; Tirado et al., 2003), and from 1100 to 2500 g in Atlantic populations (Sieiro et al., 2014; Tirado et al., 2003; Lourenço et al., 2012; Otero et al., 2007; Carvalho and Reis, 2009; Silva et al., 2002). Additionally, in Atlantic populations from Portugal and Spain (Andalusian and Galician coasts), the percentage of mature females increases steadily in the weight interval 1500 – 3000 g (Sieiro et al., 2014; Lourenço et al., 2012; Silva et al., 2002). These data are consistent with our results from Canary Islands animals reared in captivity (Fig. 2), since the majority of females showed a steep increase in *GSI* in the weight range of 2000 – 3000 g, although a smaller percentage of octopuses underwent a similar rapid *GSI* change at body weights below 2000 g, or above 4000 g. Thus, we cannot rule out that a minimal body weight is necessary before gonadal and oocyte maturation can start, but once initiated, it follows a predictable course.

Present data support a statistically significant and negative relationship between the mean *GSI* of a female in a given period, and its *SGR* in the same time interval, in spite of the fact that inter-individual variability is also apparent. O'dor and Wells (1978) reported reduced protein synthesis in arm muscles, and fast ovary growth in females entering precocious maturity after removing brain sub-pedunculate lobes. This process could affect the specific growth rate in terms of body weight. Much more recent works (González et al., 2011; Otero et al., 2007; Rosa et al., 2004) sampled wild populations of *O. vulgaris* and looked for correlations between the condition of ovary, and digestive gland or body muscles, as evidence of a trade-off between somatic and reproductive growth, but no such correlations were found. On the other hand, some authors (Sieiro et al., 2020) do not completely rule out a partial reallocation of energy from somatic to reproductive growth. In any case, with or without energy reallocation, it is not clear how ovary growth can depress the *SGR* in terms of the whole body which includes the gonad. Further studies in this area could elucidate the origin of the nutrients

that comprise the yolk, which might contribute to the enhancement of spawning quality and other related aspects.

5. Conclusion

In conclusion, these data show that the relationship between *GSIU* and spawning time can be applied to predict the start of egg laying in order to schedule future tasks and manage resources in experimental and production hatcheries. The relationship between time to spawning and *GSIU* can be particularly useful for octopus farmers since, in contrast to fish, octopus females can store spermatophores in the oviduct glands until they mature and gonad maturation does not show a clear correlation with the female weight.

Funding

This study has received funding from the European Maritime, Fisheries and Aquaculture Fund; ECOPHYN Project (PID2021-126824NB-C31-C33) funded by Spanish Ministerio de Ciencia e Innovación MICIU/AEI/10.13039/501100011033 and “ERDF/EU” and is part of the ThinkInAzul programme supported by MICIU, European Union Next-Generation EU (PRTR-C17.11) and Gobierno de Canarias (SD2218/6897). ES was funded by KU Leuven grant C14/21/065 and FWO grant G040124N.

Author statement

The manuscript entitled “*Ultrasound-based prediction of spawning time in females of Octopus vulgaris*” has not been published previously. It is not under consideration for publication elsewhere, and it has been approved by all authors and responsible authorities. The authors declare that, if it is accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright holder. The manuscript has been submitted to bioRxiv server as a preprint.

CRediT authorship contribution statement

Beatriz C. Felipe: Writing – review & editing, Investigation. **Sofia Maccuro:** Writing – review & editing, Conceptualization. **Marta Serrano:** Writing – review & editing, Investigation. **Lorenzo Márquez:** Writing – original draft, Visualization, Formal analysis, Conceptualization. **Pauline Weterings:** Writing – review & editing, Investigation, Formal analysis. **Samar Saber:** Writing – review & editing, Investigation, Conceptualization. **Eve Seuntjens:** Writing – review & editing, Supervision, Conceptualization. **Francisco Acosta-Cifuentes:** Writing – review & editing, Investigation. **María González:** Writing – review & editing. **Eduardo Almansa:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors wish to acknowledge Patrick Dennis for his assistance with the English language editing of the final manuscript.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.aqrep.2025.103025](https://doi.org/10.1016/j.aqrep.2025.103025).

Data availability

Data will be made available on request.

References

- Almansa, E., Márquez, L., Rosas, C., Martín, M.V., Navarro, J.C., Uriarte, I., Gestal, C., Fernández-Álvarez, F.A., Gallardo, P., Varó, I., Farías, A., Cardenete, G., Caamal-Monsreal, C., Rodríguez-Barreto, D. and Morales, A.E., In press. Octopods aquaculture: Reproduction, rearing technology, nutritional physiology, welfare and health status. In "Aquaculture and Living Resource Management – Volume 2", edited by Prof. Leonel Pereira to be published by CRC Press (Taylor & Francis Group). ISBN 978-1-032-34632-8.
- Berrueta, M., López, A.V., Radonic, M., Goroso, B.G., Aristizabal, E., 2025. In vivo determination of sex and reproductive status of *Octopus tewelchus* (Cephalopoda: Octopodidae) by ultrasound techniques. Mar. Fish. Sci. 38 (3), 1. <https://doi.org/10.47193/mafis.3832025010702>.
- Boles, S.E., Rofers-Bennett, L., Bragg, W.K., Bredvik-Curran, J., Graham, S., Gross, J.A., 2023. Determination of gonad reproductive state using non-lethal ultrasonography in endangered black (*Halotis cracherodii*) and white abalone (*H. sorenseni*). Front. Mar. Sci. 10, 1134844. <https://doi.org/10.3389/fmars.2023.1134844>.
- Bryant, J.L., Wildhaber, M.L., Papoulias, D.M., DeLonay, D.J., Tillitt, D.E., Annis, M.L., 2007. Estimation of gonad volume, fecundity, and reproductive stage of shovelnose sturgeon using sonography and endoscopy with application to the endangered pallid sturgeon. J. Appl. Ichthyol. 23, 411–419. <https://doi.org/10.1111/j.1439-0426.2007.00889.x>.
- Bureau du Colombier, S., Jacobs, L., Gesset, C., Elie, P., Lambert, P., 2015. Ultrasonography as a non-invasive tool for sex determination and maturation monitoring in silver eels. Fish. Res. 164, 50–58. <https://doi.org/10.1016/j.fishres.2014.10.015>.
- Carvalho, J.M.N., Reis, C.S., 2009. Contributions to knowledge on the maturation and fertility of the common octopus *Octopus vulgaris* Cuvier, 1797 on the Portuguese coast. Bol. Inst. Esp. Oceano 19, 473–481.
- Cooke, G.M., Anderson, D.B., Begout, M.L., Dennison, N., Osorio, D., Tonkins, B., Kristiansen, T., Fiorito, G., Galligioni, V., Ponte, G., Andrews, P.L.R., 2019. Prospective severity classification of scientific procedures in cephalopods: report of a COST FA1301 working group survey. Lab. Anim. 53, 541–563. <https://doi.org/10.1177/0023677219864626>.
- Coward, K., Bromage, N.R., 2001. Stereological validation of ovarian biopsy as a means of investigating ovarian condition in broodstock tilapia *in vivo*. Aquaculture 195, 183–188. [https://doi.org/10.1016/S0044-8486\(00\)00544-5](https://doi.org/10.1016/S0044-8486(00)00544-5).
- Cuccu, D., Mereu, M., Cau, A., Pesci, P., Cau, A., 2013b. Reproductive development versus estimated age and size in a wild Mediterranean population of *Octopus vulgaris* (Cephalopoda: Octopodidae). J. Mar. Biol. Assoc. U. K. 93, 843–849. <https://doi.org/10.1017/S0025315412000203>.
- Cuccu, D., Mereu, M., Porcu, C., Follesa, M.C., Cau, A.L., Cau, A., 2013a. Development of sexual organs and fecundity in *Octopus vulgaris* Cuvier, 1797 from the Sardinian waters (Mediterranean Sea). Mediterr. Mar. Sci. 14, 270–277. <https://doi.org/10.12681/mms.412>.
- Dasgupta, S., Biswas, G., Tripathi, G., Pathan, A.G., Ghoshal, T.K., Singh, P.K., Jana, T., Mitra, P., Samanta, S., De, D., Adhikari, S., Manna, R.K., Sahoo, A.K., 2024. Assessment of sex, gonad volume, and reproductive maturation status of the Indian shad, *Tenulosa ilisha*, using ultrasonic imaging: a rapid and non-invasive tool. J. Appl. Ichthyol. 2024, 6597804. <https://doi.org/10.1155/2024/6597804>.
- Di, Cristo, 2013. Nervous control of reproduction in *Octopus vulgaris*: a new model. Invert. Neurosci. 12, 27–34. <https://doi.org/10.1007/s10158-013-0149-x>.
- Di Cosmo, A., Di Cristo, C., Paolucci, M., 2001. Sex steroid hormone fluctuations and morphological changes of the reproductive system of the female of *Octopus vulgaris*. Annu. Cycle J. Exp. Zool. 289, 33–47. [https://doi.org/10.1002/1097-010X\(20010101/31\)289:1<33::AID-JEZA>3.0.CO;2-A](https://doi.org/10.1002/1097-010X(20010101/31)289:1<33::AID-JEZA>3.0.CO;2-A).
- Du, H., Zhang, X., Leng, X., Zhang, S., Luo, J., Liu, Z., Qiao, X., Kynard, B., Wei, Q., 2017. Gender and gonadal maturity stage identification of captive Chinese sturgeon, *Acipenser sinensis*, using ultrasound imagery and sex steroids. Gen. Comp. Endocrinol. 245, 36–43. <https://doi.org/10.1016/j.ygcen.2016.08.004>.
- Evans, A.F., Fitzpatrick, M.S., Siddens, L.K., 2004. Use of ultrasound imaging and steroid concentrations to identify maturational status in adult steelhead. N. Am. J. Fish. Manag. 24, 967–978. <https://doi.org/10.1577/M03-112.1>.
- Fiorito, G., Affuso, A., Basil, J., Cole, A., de Girolamo, P., D'Angelo, L., Dickel, L., Gestal, C., Grasso, F., Kuba, M., Mark, F., Melillo, D., Osorio, D., Perkins, K., Ponte, G., Shashar, N., Smith, D., Smith, J., Andrews, P.L.R., 2015. Guidelines for the care and welfare of cephalopods in research – a consensus based on an initiative by CephRes, FELASA and the Boyd Group. Lab. Anim. 49, 1–90. <https://doi.org/10.1177/0023677215580006>.
- Forsythe, J.W., Hanlon, R.T., 1988. Effect of temperature on laboratory growth, reproduction and life span of *Octopus bimaculoides*. Mar. Biol. 98, 369–379. <https://doi.org/10.1007/BF00391113>.
- González, M., Barcala, E., Pérez-Gil, J.L., Carrasco, M.N., García-Martínez, M.C., 2011. Fisheries and reproductive biology of *Octopus vulgaris* (Mollusca: Cephalopoda) in the Gulf of Alicante (Northwestern Mediterranean). Medit. Mar. Sci. 12, 369–389. <https://doi.org/10.12681/mms.38>.
- Grimaldi, A.M., Agnola, G., Fiorito, G., 2007. Using ultrasound to estimate brain size in the cephalopod *Octopus vulgaris* Cuvier *in vivo*. Brain Res. 1183, 66–73. <https://doi.org/10.1016/j.brainres.2007.09.032>.
- Jennings, C.A., Will, T.A., Reinert, T.R., 2005. Efficacy of a high- and low-frequency ultrasonic probe for measuring ovary volume and estimating fecundity of striped bass *Morone saxatilis* in the Savannah River Estuary. Fish. Res. 76, 445–453. <https://doi.org/10.1016/j.fishres.2005.07.016>.
- Katsanevakis, S., Verriopoulos, G., 2006. Seasonal population dynamics of *Octopus vulgaris* in the eastern Mediterranean. ICES J. Mar. Sci. 63, 151–160. <https://doi.org/10.1016/j.icesjms.2005.07.004>.
- Lourenço, S., Moreno, A., Narciso, L., González, A.F., Pereira, J., 2012. Seasonal trends of the reproductive cycle of *Octopus vulgaris* in two environmentally distinct coastal areas. Fish. Res. 127–128, 116–124. <https://doi.org/10.1016/j.fishres.2012.04.006>.
- Macbeth, B.J., Frimer, H.D., Muscatello, J.R., Janz, D.M., 2011. Use of portable ultrasonography to determine ovary size and fecundity non-lethally in northern pike (*Esox lucius*) and white sucker (*Catostomus commersoni*). Water Qual. Res. J. Can. 46, 43–51. <https://doi.org/10.2166/wqrjc.2011.011>.
- Mangold, K., 1983. Food, feeding and growth in cephalopods. Mem. Natl. Mus. Vic. 44, 81–93. <https://doi.org/10.24199/j.mmv.1983.44.08>.
- Margheri, L., Ponte, G., Mazzolai, B., Laschi, C., Fiorito, G., 2011. Non-invasive study of *Octopus vulgaris* arm morphology using ultrasound. J. Exp. Biol. 214, 3727–3731. <https://doi.org/10.1242/jeb.057323>.
- Márquez, L., Martín, M.V., Larson, M., Almansa, E., 2023. Application of the thermal time model of embryonic development duration to the culture of octopod cephalopods. Aquac. Rep. 30, 101563. <https://doi.org/10.1016/j.aqrep.2023.101563>.
- Mattson, N.S., 1991. A new technique to determine sex and gonad size in live fishes by using ultrasonography. J. Fish. Biol. 39, 673–677. <https://doi.org/10.1111/j.1095-8649.1991.tb04397.x>.
- McGarvey, L.M., Ilgen, J.E., Guy, C.S., McLellan, J.G., Webb, M.A.H., 2021. Gonad size measured by ultrasound to assign stage of maturity in burbot. J. Fish. Wildl. Manag. 12, 241–249. <https://doi.org/10.3996/JFWM-20-082>.
- Medit Working Group, 2017. International bottom trawl survey in the Mediterranean. Instruction manual. Version 9.
- Minakata, H., Tsutsui, K., 2016. Oct-GnRH, the first protostomian gonadotropin-releasing hormone-like peptide and a critical mini-review of the presence of vertebrate sex steroids in molluscs. Gen. Comp. Endocrinol. 227, 109–114. <https://doi.org/10.1016/j.ygcen.2015.07.011>.
- Mlingi, F.T., Puvanendran, V., Burgerhout, E., Tveiten, H., Tomkiewicz, J., Kjorsvik, E., Mommens, M., 2023. Ultrasonic imaging as a means of monitoring gonadal development in lumpfish (*Cyclopterus lumpus*). e15811 Physiol. Rep. 11. <https://doi.org/10.14814/phyt2.15811>.
- Naeve, I., Mommens, M., Arukwe, A., Kjorsvik, E., 2018. Ultrasound as a noninvasive tool for monitoring reproductive physiology in female Atlantic salmon (*Salmo salar*). Physiol. Rep. 6, e13640. <https://doi.org/10.14814/phyt2.13640>.
- Novelo, N.D., Tiersch, T.R., 2012. A review of the use of ultrasonography in fish reproduction. N. Am. J. Aquacult. 74, 169–181. <https://doi.org/10.1080/15222055.2012.672370>.
- O'dor, R.K., Wells, M.J., 1978. Reproduction versus somatic growth: hormonal control in *Octopus vulgaris*. J. Exp. Biol. 77, 15–31. <https://doi.org/10.1242/jeb.77.1.15>.
- Otero, J., González, A.F., Sieiro, M.P., Guerra, A., 2007. Reproductive cycle and energy allocation of *Octopus vulgaris* in Galician waters, NE Atlantic. Fish. Res. 85, 122–129. <https://doi.org/10.1016/j.fishres.2007.01.007>.
- Polese, G., Bertapelle, C., Di Cosmo, A., 2015. Role of olfaction in *Octopus vulgaris* reproduction. Gen. Comp. Endocrinol. 210, 55–62. <https://doi.org/10.1016/j.ygcen.2014.10.006>.
- Ponte, G., Sykes, A.V., Cooke, G.M., Almansa, E., Andrews, P.R.L., 2017. The digestive tract of cephalopods: toward non-invasive *in vivo* monitoring of its physiology. Front. Physiol. 8, 403. <https://doi.org/10.3389/fphys.2017.00403>.
- Reis, D., Shcherbakova, A., Riera, R., Martín, V.M., Domingues, P., Andrade, J.P., Jiménez-Prada, P., Rodríguez, C., Sykes, A.V., Almansa, E., 2021. Effects of feeding with different live preys on the lipid composition, growth and survival of *Octopus vulgaris* paralarvae. Aquac. Res. 52, 105–116. <https://doi.org/10.1111/are.14873>.
- Rosa, R., Costa, P.R., Nunes, M.L., 2004. Effect of sexual maturation on the tissue biochemical composition of *Octopus vulgaris* and *O. defilippi* (Mollusca: Cephalopoda). Mar. Biol. 145, 563–574. <https://doi.org/10.1007/s00227-004-1340-8>.
- Saber, S., Macías, D., Ortiz de Urbina, J., Kjesbu, O.S., 2015. Stereological comparison of oocyte recruitment and batch fecundity estimates from paraffin and resin sections using spawning albacore (*Thunnus alalunga*) ovaries as a case study. J. Sea Res. 95, 226–238. <https://doi.org/10.1016/j.fishres.2014.05.003>.
- Sánchez, F.J., Cerezo, J., García, B., 2014. *Octopus vulgaris*: on-growing. In: Iglesias, J., Fuentes, L., Villanueva, R. (Eds.), Cephalopod Culture. Springer, New York, pp. 451–466. https://doi.org/10.1007/978-94-017-8648-5_24.
- Scherle, W., 1970. A simple method for volumetry of organs in quantitative stereology. Mikroskopie 26, 57–60.
- Sieiro, P., Otero, J., Aubourg, S.P., 2020. Biochemical composition and energy strategy along the reproductive cycle of female *Octopus vulgaris* in Galician waters (NW Spain). Front. Physiol. 11, 760. <https://doi.org/10.3389/fphys.2020.00760>.
- Sieiro, P., Otero, J., Guerra, A., 2014. Contrasting macroscopic maturity staging with histological characteristics of the gonads in female *Octopus vulgaris*. Hydrobiologia 730, 113–125. <https://doi.org/10.1007/s10750-014-1826-4>.
- Silva, L., Sobrino, I., Ramos, F., 2002. Reproductive biology of the common octopus *Octopus vulgaris* Cuvier, 1797 (Cephalopoda: Octopodidae) in the Gulf of Cádiz (SW Spain). Bull. Mar. Sci. 71, 837–850.
- Tirado, C., de la Rúa, Rodríguez, Bruzón, A., López, M.A., Márquez, I., J.I., 2003. La reproducción del pulpo (*Octopus vulgaris*) y el choco (*Sepia officinalis*) en la costa andaluza. Cons. fa De. Agric. Y Pesca Junta De. Andal. fa. Sevilla.

- Wang, Z.Y., Pergande, M.R., Ragsdale, C.W., Cologna, S.M., 2022. Steroid hormones of the octopus self-destruct system. *Curr. Biol.* 32, 2572–2579. <https://doi.org/10.1016/j.cub.2022.04.043>.
- Wells, M.J., Wells, J., 1959. Hormonal control of sexual maturity in Octopus. *J. Exp. Biol.* 36, 1–33. <https://doi.org/10.1242/jeb.36.1.1>.
- Whiteman, E.A., Jennings, C.A., Nemeth, R.S., 2005. Sex structure and potential female fecundity in a *Epinephelus guttatus* spawning aggregation: applying ultrasonic imaging. *J. Fish. Biol.* 66, 983–995. <https://doi.org/10.1111/j.1095-8649.2005.00653.x>.
- Yari, Y., Næve, I., Hammerdal, A., Bergtun, P.H., Måsøy, S.E., Voormolen, M.M., Lovstakken, L., 2024. Automated measurement of ovary development in Atlantic salmon using deep learning. *Ultrasound Med. Biol.* 50, 364–373. <https://doi.org/10.1016/j.ultrasmedbio.2023.11.008>.
- Zou, W., Hong, J., Ma, Y., Yu, W., Liu, Y., Ai, C., Huang, M., Luo, X., You, W., Ke, C., 2025. Application of ultrasonography in abalone gonadal evaluation (In press.). *Aquac. Fish.* <https://doi.org/10.1016/j.aaf.2024.04.001>.